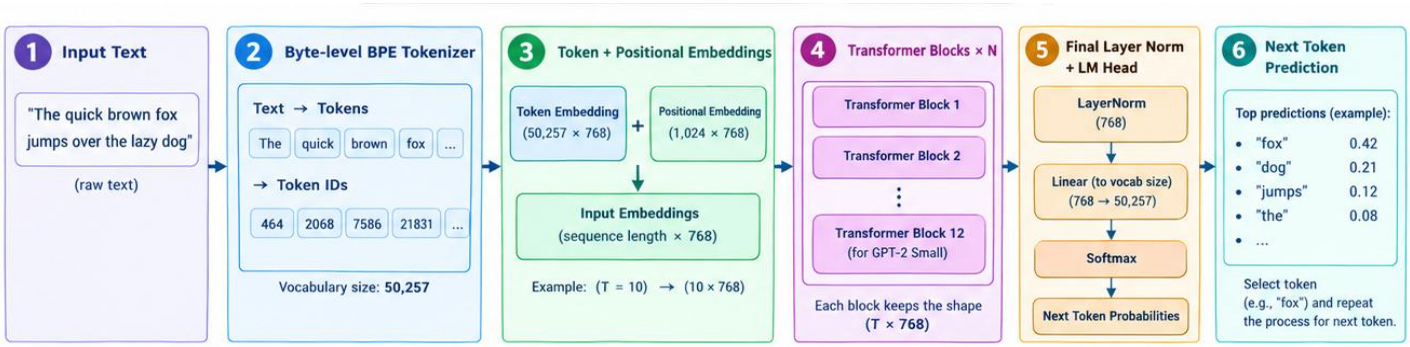
GPT-2 ARCHITECTURE

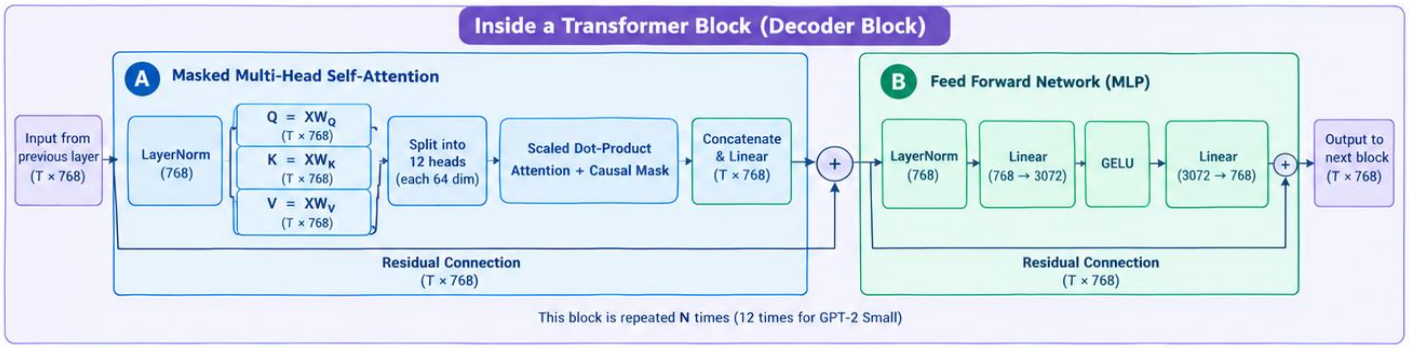
ISHA TYAGI

20240110040010O

CSE-AIML, B

5th Semester





Model Configurations

Size	Parameters	Layers	d_model (embedding dim)	Heads	MLP dim	Context Length
Small	124M	12	768	12	3,072	1,024
Medium	355M	24	1,024	16	4,096	1,024
Large	774M	36	1,280	20	5,120	1,024
XL	1.5B	48	1,600	25	6,400	1,024

Dataset & Training

Dataset: WebText
(≈ 8M documents, 40 GB, English web text)

Tokenizer: Byte-level BPE
(Vocab size: 50,257)

Training Objective:
Next-token prediction
(Cross-entropy loss)

Optimizer: AdamW
(Batch size: 512, LR tuned per run)

Key Dimensions (GPT-2 Small)

Vocabulary size (V)	50,257
Max sequence length (T)	1,024
Embedding dimension (d_{model})	768
Attention heads	12
Head dimension (d_k)	64
MLP hidden dimension	3,072
Transformer blocks	12
Total parameters	124M

Important Notes

- GPT-2 is a decoder-only Transformer (no encoder).
- Uses causal (masked) self-attention to prevent looking at future tokens.
- Positional embeddings are learned (not sinusoidal).
- Token embedding and LM head weights are tied (shared).
- Trained on a large-scale internet dataset (WebText) for general language understanding and generation.